Gradient Descent and Automatic Differentiation

Computer Seminar — Week 3 (80 minutes)

DSBA

Plan and timing

Activity	Minutes
Lecture recap and setup	8
Fixed-step GD and stability	15
Conditioning and trajectories	12
Backtracking and stopping	10
Manual graph and scalar AD	22
Gradient checking and model training	13

Recap: five formulas

$$x_{k+1} = x_k - \alpha \nabla f(x_k), \qquad 0 < \alpha < 2/L,$$

$$f(x - \alpha \nabla f(x)) \leq f(x) - \alpha(1 - L\alpha/2) \|\nabla f(x)\|^2,$$

$$f(x + \alpha d) \leq f(x) + c\alpha \nabla f(x)^\top d,$$

$$f(x_k) - f^* = O(1/k) \quad \text{(convex)},$$

$$f(x_k) - f^* \leq (1 - \mu/L)^k (f(x_0) - f^*) \quad \text{(PL)}.$$

Notebook conventions

Use only NumPy/Matplotlib for the optimizer and scalar AD. SciPy or scikit-learn may be used only as a final reference check. Every experiment must report the objective, gradient norm, accepted step, and iteration count. Fix random seeds.

Task 1: fixed-step gradient descent

Implement `gradient_descent(f, grad, x0, step, ...)` returning a history dictionary. Stop using a gradient criterion plus at least one safeguard. Test on

$$f(x) = \tfrac{1}{2} x^\top Q x - b^\top x.$$

Do not use an optimizer library.

Task 2: find the stability boundary

Use $Q = \mathrm{diag}(1, 10)$ and compare

$$\alpha \in \{0.05, 0.19, 0.20, 0.21\}.$$

Plot $f(x_k) - f^*$ on a log scale. Explain oscillation at $2/L$ and divergence above it using the eigen-coordinate recurrence.

Task 3: conditioning changes trajectories

Compare $Q_1 = \mathrm{diag}(1, 2)$ and $Q_2 = \mathrm{diag}(1, 50)$ from the same x_0. Overlay contour lines and iterates. Report κ, the observed iteration count, and the predicted contraction factor.

Task 4: Armijo backtracking

Implement line search with $\alpha_0 = 1$, $\beta = 0.5$, $c = 10^{-4}$. Record the number of function evaluations and accepted steps. Compare with a safe constant step on the poorly conditioned quadratic.

Guardrails

Reject non-descent directions; cap reductions; do not recompute the gradient inside the line-search loop.

Task 5: stopping is a decision

Run the same problem under three single criteria: gradient norm, step norm, and function change. Construct or identify a case where a change-based test stops prematurely. Then implement a combined rule and state its scaling convention.

Task 6: a graph by hand

For $z = \sin(xy) + x^2$ at $(x, y) = (2, -1)$:

1. list graph nodes in topological order;
2. compute every forward value;
3. propagate adjoints backward;
4. verify $\partial z/\partial x$ and $\partial z/\partial y$ analytically.

Task 7: scalar automatic differentiation

Complete a `Value` class supporting `+`, `*`, powers, negation, subtraction, division, `exp`, `log`, and `tanh`. Each result stores parents and a local backward closure.

Reverse pass: essential invariant

Build a topological ordering by DFS, seed `output.grad = 1`, then visit nodes in reverse order. Local rules must use +=: shared subexpressions create multiple gradient paths.

Task 8: gradient checking

Use central differences

$$\frac{\partial f}{\partial x_i} \approx \frac{f(x + he_i) - f(x - he_i)}{2h}.$$

Compare with AD over $h = 10^{-1}, \ldots, 10^{-10}$. Plot relative error; explain truncation error on the left and round-off error on the right.

Task 9: train a model with your optimizer

Generate a binary dataset. Implement stable logistic loss and its gradient, train with your fixed-step GD and with Armijo, and compare histories. Report accuracy only after analyzing the optimization objective and gradient norm.

Optional integration: train through scalar AD

For a small subset, express the logistic loss entirely with `Value` objects. Compare its parameter gradients with the vectorized analytic gradient. Explain why scalar AD is educational but slow.

Expected deliverables

- completed functions with concise docstrings and failure checks;
- four diagnostic figures with labeled axes and legends;
- a table comparing step strategies and conditioning;
- two short written explanations: stability and gradient accumulation;
- reproducible execution from a fresh kernel.

Exit questions

Why is $-\nabla f$ geometry-dependent? Why can $\alpha = 2/L$ fail? What does Armijo certify? Where does reverse mode save work? Why must adjoints accumulate?